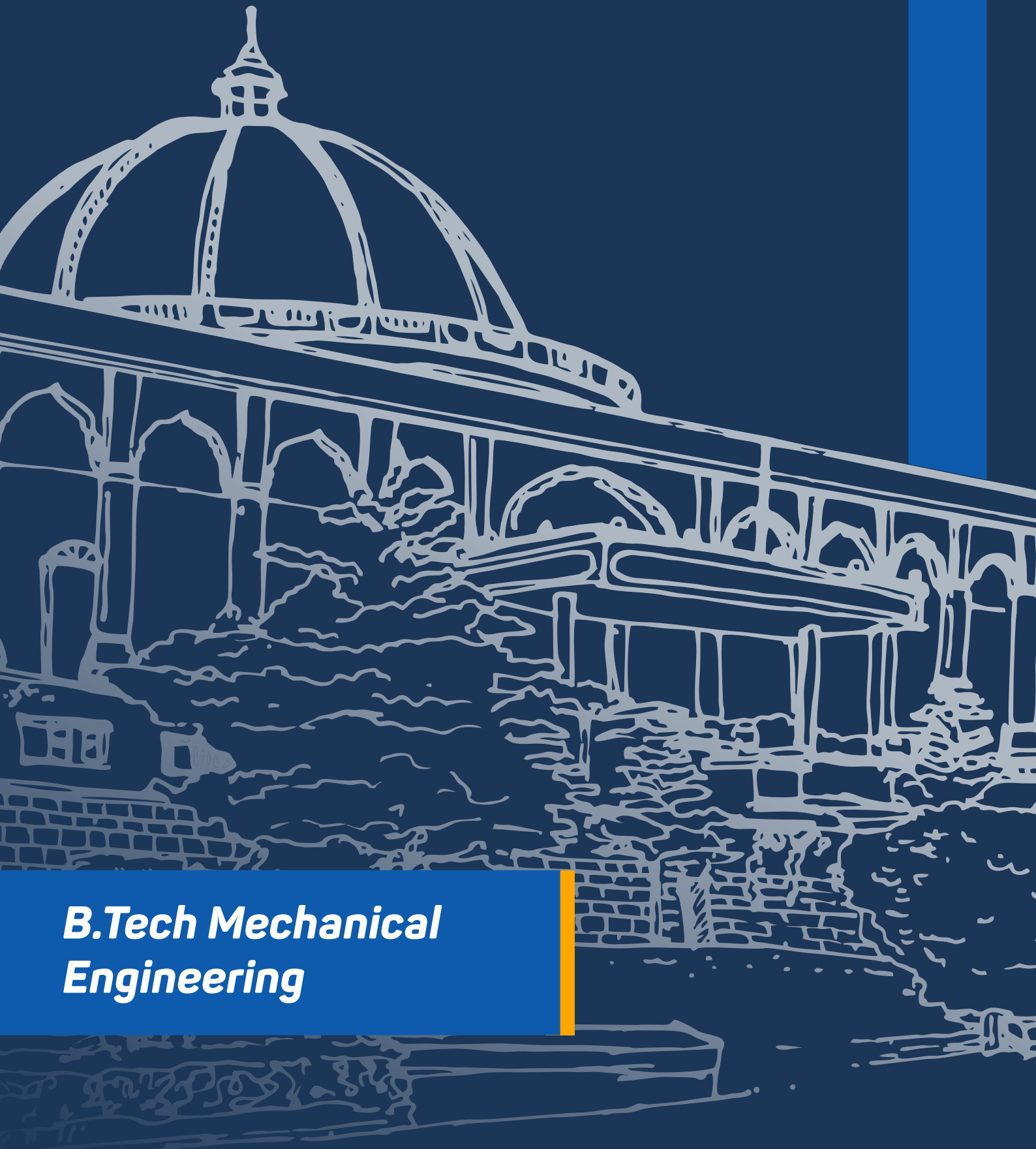




B.Tech Mechanical Engineering

Division	Faculty of Engineering and Technology
School Name	School of Engineering & Technology
Department Name	Department of Mechanical Engineering
Programme Name	B.Tech Mechanical Engineering

Semester	Odd (I)	Even (II)	Total Credits
First Year	22	21	43
Second Year	22	22	44
Third Year	22	22	44
Fourth Year	15	19	34

+ + + + + + + + + + + + + + + **COURSE BASKET** + + + + + + + + + + + + + + +

| Course Type | Description |
|----------------------|--|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU |
| Programme Electives | Open electives under the programme allow students to specialise in a particular area connected to their major. |
| University Core | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core. |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---------------------------------------|---------------|
| I | PF | Complex Variable and Linear Algebra | 3 |
| I | PF | Engineering Physics | 3 |
| I | PF | Engineering Chemistry | 3 |
| I | PF | Biology for Engineers | 2 |
| I | PF | Engineering Graphics | 3 |
| I | PF | Programming and Problem Solving | 3 |
| I | UC | Indian Constitution | 1 |
| I | UC | Environment and Sustainability | 1 |
| I | UC | Yoga - I | 1 |
| I | UC | Social Leadership Development Program | 1 |
| I | UC | Financial Literacy | 1 |
| | | Total Credits: | 22 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| II | PF | Calculus | 3 |
| II | PF | Basics of Electrical and Electronics Engineering | 3 |
| II | PF | Engineering Mechanics | 3 |
| II | Program Capstone Project/Seminar and Internships | Computer Aided Modelling | 2 |
| II | PF | Ideas and Innovations in Manufacturing | 1 |
| II | UC | Yoga - II | 1 |
| II | UC | Co-creation | 1 |
| II | UC | AI for everyone | 2 |
| II | UC | Foundation of Peace | 2 |
| II | UC | Indian Knowledge System (General) | 2 |
| II | UC | Sports | 1 |
| | | Total Credits: | 21 |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| III | PF | Differential Equations and Transform Techniques | 4 |
| III | PM | Engineering Thermodynamics | 3 |
| III | PM | Strength of Materials | 3 |
| III | Program Capstone Project/
Seminar and Internships | Design Thinking | 1 |
| III | PF | Engineering Metallurgy and Materials Science | 3 |
| III | PF | Data Science and Statistics | 2 |
| III | UE | University Electives - I | 3 |
| III | UC | Spiritual and Cultural heritage: Indian Experience | 2 |
| III | UC | Research Innovation Design Entrepreneurship (RIDE) | 1 |
| | | Total Credits: | 22 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| IV | PM | Fluid Mechanics & Machines | 4 |
| IV | PM | Theory of Machines | 3 |
| IV | PF | Indian knowledge System-II (Sci & Tech) | 2 |
| IV | PM | Manufacturing Science | 4 |
| IV | PM | Design of Machine Elements | 3 |
| IV | Program Capstone Project/
Seminar and Internships | Mechanical Engineering Software Lab | 1 |
| IV | UE | University Electives - II | 3 |
| IV | UC | Rural Immersion | 1 |
| IV | UC | Life Transformation Skills | 1 |
| | | Total Credits: | 22 |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| V | PM | Heat Transfer | 3 |
| V | PM | Transmission System Design | 4 |
| V | PM | Mechatronics and Control System | 3 |
| V | PF | Metrology and Quality Control | 3 |
| V | PE | Program Elective - I | 4 |
| V | UE | University Electives - III | 3 |
| V | UC | Managing Conflicts Peacefully: Tools and Techniques | 2 |
| | | Total Credits: | 22 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| VI | PM | Applied Thermal Engineering | 4 |
| VI | PM | Numerical Methods | 3 |
| VI | PM | Dynamics of Machinery | 4 |
| VI | PM | AI in Mechanical Engineering | 2 |
| VI | PM | Project Management and Operation Research | 2 |
| VI | PE | Program Elective - II | 4 |
| VI | Program Capstone Project/Seminar and Internships | Mini Project Lab | 1 |
| VI | UC | National Academic Immersion Program | 2 |
| | | Total Credits: | 22 |

Semester - VII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|----------------------------|---------------|
| VII | Program Capstone Project/Seminar and Internships | Capstone Project Stage I | 1 |
| VII | Program Capstone Project/Seminar and Internships | Internship | 14 |
| | | Total Credits: | 15 |

Semester - VIII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|----------------------------|---------------|
| VIII | PE | Program Elective - III | 4 |
| VIII | PE | Program Elective - IV | 4 |
| VIII | Program Capstone Project/Seminar and Internships | Capstone Project Stage II | 8 |
| VIII | PM | Power Engineering | 3 |
| | | Total Credits: | 19 |

Program Electives

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---|---------------|
| V | PE - I | Finite Element Analysis | 4 |
| V | PE - I | Hydraulics and Pneumatics | 4 |
| V | PE - I | Additive and Digital Manufacturing | 4 |
| V | PE - I | Energy Storage and Management for Electric Vehicles | 4 |
| VI | PE - II | Acoustics and Noise Control | 4 |
| VI | PE - II | Refrigeration and Air-Conditioning | 4 |
| VI | PE - II | Manufacturing Planning and Control | 4 |
| VI | PE - II | Advanced Materials | 4 |
| VIII | PE - III | Mechanical System Design and Modelling | 4 |
| VIII | PE - III | Automobile and E - Vehicle Technology | 4 |
| VIII | PE - III | Computer Integrated Manufacturing | 4 |
| VIII | PE - III | Product Design and Development | 4 |
| VIII | PE - IV | Design and Optimization of Experiments | 4 |
| VIII | PE - IV | Computational Fluid Dynamics | 4 |
| VIII | PE - IV | Industrial Engineering and Management | 4 |
| VIII | PE - IV | Industrial Robotics and Programming | 4 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.